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HSSC-I NATIONAL BOOK FOUNDATION FBISE

CHEMISTRY TEST - CHAPTER 22: ENERGY

Name:		Roll No.		Section:

Total Time: 2:30 Hour
Time Allowed: 150 Minutes

Total Marks: 150

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. Coal, oil and natural gas are called:	biofuels	renewable fuels	fossil fuels	nuclear fuels	○○●○
2. Fossil fuels are formed from the remains of:	metals	rocks only	organisms	water only	○○●○
3. The word "petroleum" comes from the Latin words meaning:	water and gas	rock and oil	earth and fire	coal and oil	○●○○
4. Petra in the word petroleum means:	stone or rock	gas	water	oil	●○○○
5. Petroleum is also called:	crude oil	natural gas	coal tar	kerosene	●○○○
6. Crude oil is a thick, dark liquid composed mainly of:	carbohydrates	metals	hydrocarbons	proteins	○○●○
7. Natural gas consists mostly of:	propane	methane	ethane	butane	○●○○
8. Besides methane, natural gas also contains significant amounts of ethane, propane and:	ethanol	benzene	butane	glycerol	○○●○
9. Petroleum is believed to have formed from:	marine plants and animals that lived hundreds of millions of years ago	desert sand	volcanic rocks	ice	●○○○
10. Petroleum forms from buried organic matter due to heat, pressure and:	wind	presence of oxygen	sunlight	lack of air and bacterial activity	○○○●
11. The gas that accumulates on top of oil deposits is called:	natural gas	coal gas	water gas	producer gas	●○○○
12. Crude oil is extracted from the Earth's crust by:	filtering rocks	melting rocks	drilling holes	mining only	○○●○
13. When drilling for oil, the substance that first comes out at high pressure is often:	coal	crude oil	natural gas	water	○○●○
14. The conversion of crude oil into useful products is called:	reforming	cracking	refining	polymerisation	○○●○

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Questions	A	B	C	D	A B C D
15. Crude oil is separated into fractions by:	crystallisation	fractional distillation	filtration	evaporation only	○●○○
16. The separation of crude oil in a fractionating column is based on differences in:	colours	boiling points	molar volumes	densities only	○●○○
17. In a fractionating column, compounds with the highest boiling points condense:	in the middle	near the bottom	do not condense	near the top	○●○○
18. In a fractionating column, compounds with the lowest boiling points condense:	in the middle	near the top	do not condense	near the bottom	○●○○
19. Compounds that do not boil in the fractionating column collect at the bottom as:	residue	kerosene	gas	naphtha	●○○○
20. Each fraction of petroleum has its own specific:	boiling range and composition	smell only	taste	colour only	●○○○
21. The petroleum gas fraction contains hydrocarbons with how many carbon atoms?	10 to 16	5 to 10	1 to 4	20 to 50	○○●○
22. The petroleum gas fraction boils below:	350 °C	170 °C	70 °C	20 °C	○○○●
23. The petroleum gas fraction is used as:	jet fuel	cylinder gas for cooking	lubricating oil	road surfacing	○●○○
24. LPG stands for:	light petroleum gas	liquid pure gas	liquefied petroleum gas	low pressure gas	○○●○
25. The naphtha fraction boils between about:	170–250 °C	70–120 °C	500–600 °C	350–500 °C	○●○○
26. Naphtha is used in making:	ships' fuel	road surfaces	drugs and plastics	candles only	○○●○
27. The kerosene fraction boils between about:	70–120 °C	170–250 °C	350–500 °C	below 20 °C	○●○○
28. Kerosene is used for:	lubrication	road surfacing	cooking gas	heating, lighting and aircraft fuel	○○○●
29. The lubricating oil fraction contains about how many carbon atoms?	more than 70	1 to 4	8 to 12	20 to 50	○○○●
30. The lubricating oil fraction boils between about:	350–500 °C	below 20 °C	170–250 °C	70–120 °C	●○○○
31. Lubricating oil is used in:	cooking	machines and engines, and for waxes	plastics only	aeroplane fuel only	○●○○

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Questions	A	B	C	D	A B C D
32. Fuel oil is used in:	factories and ships	cars	household cooking	airplanes	●○○○
33. Fuel oil contains about how many carbon atoms?	8 to 12	50 to 70	1 to 4	20 to 50	○●○○
34. Bitumen has more than how many carbon atoms?	70	10	50	20	●○○○
35. Bitumen is used for:	cars	lighting	roofing and road surfacing	cooking	○○●○
36. Which petroleum fraction has the highest boiling point?	Kerosene	Naphtha	Fuel oil	Bitumen	○○○●
37. Which petroleum fraction has the lowest boiling point?	Naphtha	Petroleum gas	Kerosene	Bitumen	○●○○
38. Organic compounds derived from petroleum are called:	petrochemicals	minerals	proteins	carbohydrates	●○○○
39. Which of the following is a petrochemical?	Ethylene	Oxygen	Water	Sodium chloride	●○○○
40. Among these, a petrochemical is:	Sand	Salt	Iron	Benzene	○○○●
41. A petrochemical obtained from petroleum is:	Sodium	Methanol	Chlorine	Calcium	○●○○
42. Petrochemicals are mainly:	inorganic salts	noble gases	metals	hydrocarbons	○○○●
43. Petrochemicals are obtained from crude oil, gas oil and:	air	petroleum gas	metal ores	sea water	○●○○
44. Petrochemicals can also be obtained from renewable sources like:	sand and gravel	corn, sugarcane and biomass	iron ore	salt	○●○○
45. Which of the following is a petrochemical product?	Marble	Cement	Glass	Plastics	○○○●
46. Petrochemical products include rubbers, synthetic fibres, dyes, adhesives and:	glass	detergents	cement	metals	○●○○
47. The process in which high molecular weight hydrocarbons are heated at high temperature and pressure to form smaller hydrocarbons is:	distillation	catalytic cracking	steam reforming	thermal cracking	○○○●
48. Thermal cracking is carried out at about:	100–200 °C	20–50 °C	450–750 °C	1000–1500 °C	○○●○

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Questions	A	B	C	D	A B C D
49. Thermal cracking is carried out at a pressure of about:	70 atm	700 atm	10 atm	1 atm	●○○○
50. Thermal cracking produces light oil (gasoline), medium oil, heavy oil and:	ethanol	glycerol	coke	water	○○●○
51. Medium oil from thermal cracking is used as:	lubricant	cooking gas	diesel fuel	road surfacing	○○●○
52. The process in which a catalyst is used to break down large hydrocarbons is:	thermal cracking	distillation	catalytic cracking	steam reforming	○○●○
53. The catalyst used in catalytic cracking is:	platinum wire	iron	zeolites	nickel	○○●○
54. Catalytic cracking is used to produce:	coke only	bitumen	lubricating oil	high-octane fuels (petrol)	○○○●
55. Both cracking processes are important for producing:	plastics only	metals	commercial gasoline and light oils	bitumen	○○●○
56. The gaseous products of cracking such as ethene and propene are raw materials for:	cement plants	petrochemical plants	steel plants	fertiliser plants only	○●○○
57. In steam reforming, lower hydrocarbons are heated with steam in the presence of a:	solvent	catalyst	base	dehydrating agent	○●○○
58. Steam reforming produces a mixture of H_2 and:	CO	N_2	O_2	CO_2 only	●○○○
59. Steam reforming is carried out at about:	700–1000 °C	2000 °C	20–50 °C	100–200 °C	●○○○
60. The feedstocks for steam reforming include natural gas, refinery gas, LPG and:	bitumen	fuel oil	lubricating oil	low boiling naphtha	○○○●
61. CO and H_2 from steam reforming are further converted into:	benzene	ethanol	methanol	glycerol	○○●○
62. Materials that store energy and release it for use are collectively called:	fuels	fertilisers	catalysts	solvents	●○○○
63. Energy stored in a material per unit volume is called:	specific energy	power	energy density	work	○○●○
64. The unit of energy density is:	N	$J\ m^{-3}$	W	$J\ kg^{-1}$	○●○○
65. The amount of energy a fuel releases when burned per unit mass is called:	power	work	energy density	specific energy	○○○●

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Questions	A	B	C	D	A B C D
66. The unit of specific energy is:	N	J kg^{-1}	J m^{-3}	W	○●○○
67. The specific energy of coal is about:	16 MJ/kg	55 MJ/kg	46 MJ/kg	24 MJ/kg	○○○●
68. The specific energy of natural gas is about:	46 MJ/kg	24 MJ/kg	16 MJ/kg	55 MJ/kg	○○○●
69. The specific energy of gasoline is about:	16 MJ/kg	55 MJ/kg	46 MJ/kg	24 MJ/kg	○○●○
70. The specific energy of wood is about:	16 MJ/kg	46 MJ/kg	24 MJ/kg	55 MJ/kg	●○○○
71. Which fuel in the table has the highest specific energy?	Uranium-235	Coal	Gasoline	Natural gas	●○○○
72. Which of the following has the lowest specific energy?	Natural gas	Wood	Gasoline	Coal	○●○○
73. The specific energy of uranium-235 is about:	46 MJ/kg	24 MJ/kg	55 MJ/kg	3 900 000 MJ/kg	○○○●
74. Uranium can produce energy through:	combustion only	oxidation only	fission and fusion	hydrolysis	○○●○
75. The energy of uranium is stored in the:	outer shell only	nucleus of its atoms	electrons of its atoms	chemical bonds	○●○○
76. The efficiency of a battery is the energy it provides compared to the energy required to:	store it	discard it	charge it	manufacture it	○○●○
77. No battery has 100% efficiency because some energy is lost during:	storage, charging and discharging	use as a light	cooling only	manufacturing only	●○○○
78. The efficiency of lithium-ion batteries is about:	99%	10%	85%	50%	●○○○
79. The efficiency of a lead-acid battery is about:	50%	10%	99%	85%	○○○●
80. Nuclear fission occurs when a heavy nucleus:	combines with another nucleus	absorbs a proton	emits electrons only	splits into smaller nuclei	○○○●
81. Nuclear fission is usually initiated by the absorption of a:	neutron	electron	proton	alpha particle	●○○○
82. The nuclei that commonly undergo fission are uranium-235 and:	plutonium-239	hydrogen-1	carbon-12	helium-4	●○○○

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Questions	A	B	C	D	A B C D
83. In the fission of uranium-235, the number of neutrons emitted in the example is:	2	1	3	5	○○●○
84. The fission of U-235 by neutron absorption produces Kr-89, Ba-141 and:	two protons	no neutrons	one electron only	three neutrons and energy	○○○●
85. The lost mass (mass defect) in nuclear reactions is converted into energy according to:	E = hν only	PV = nRT	$E = mc^2$	F = ma	○○●○
86. In nuclear fission, the mass of the reactants is ___ the mass of the products.	equal to	unrelated to	less than	greater than	○○○●
87. Nuclear fusion is a process in which two light nuclei:	are destroyed	emit only light	combine to form a heavier nucleus	split into two nuclei	○○●○
88. Nuclear fusion requires:	extremely high temperatures	very low temperatures	no energy	room temperature	●○○○
89. In the fusion reaction in the book, deuterium and tritium fuse to form helium-4, a neutron and:	a large amount of energy	oxygen	an electron	a proton	●○○○
90. Deuterium and tritium are isotopes of:	hydrogen	lithium	carbon	helium	●○○○
91. Fusion reactions require high temperature to overcome the:	gravitational attraction	electrostatic repulsion between nuclei	magnetic force only	weak nuclear force	○●○○
92. A fusion reaction releases ___ energy than a fission reaction.	less	the same	no	more	○○○●
93. The main by-product of fusion is:	carbon dioxide	helium	radioactive waste	methane	○●○○
94. Fission produces:	only water	long-lived radioactive waste	only helium	no waste	○●○○
95. Nuclear energy can provide electricity without producing:	neutrons	CO_2 and other air pollutants	heat	any waste	○●○○
96. Nuclear power plants produce ___ atmospheric pollution than thermal power plants.	equal	infinitely more	less	more	○○●○
97. Burning fossil fuels releases large amounts of greenhouse gases such as:	oxygen	carbon dioxide	nitrogen only	helium	○●○○
98. Greenhouse gases are responsible for:	global warming	noble gas release	acid rain only	ozone formation	●○○○
99. Which of the following is an impact of global warming?	Sea level rise	More glaciers	Lower temperature	Lower sea level	●○○○

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Questions	A	B	C	D	A B C D
100. The total amount of greenhouse gases produced directly or indirectly by human activities is called the:	carbon cycle	nitrogen cycle	energy density	carbon footprint	○○○●
101. Carbon footprint is usually expressed in:	tons of CO_2 equivalent per year	volts	joules per kg	litres	●○○○
102. Which of the following is a renewable energy source?	Natural gas	Petroleum	Coal	Solar energy	○○○●
103. A renewable source of energy is:	Uranium	Coal	Wind	Oil	○○●○
104. Which of these energy sources is renewable?	Hydropower	Natural gas	Coal	Oil	●○○○
105. Which energy source below is renewable and inexhaustible?	Geothermal	Petrol	Coal	Diesel	●○○○
106. Renewable energy sources produce electricity with ___ emissions compared to fossil fuels.	more	the same	significantly less or no	double	○○●○
107. Transition to renewable energy helps to:	increase carbon footprint	stop all electricity	increase pollution	reduce air pollution and increase energy independence	○○○●
108. The energy that comes from the sun is called:	wind energy	tidal energy	geothermal energy	solar energy	○○○●
109. Solar panels are used to convert sunlight into:	only light	electricity and heat	only chemical energy	only sound	○●○○
110. The two main solar energy technologies are thermal systems and:	hydropower plants	photovoltaic cells	wind turbines	nuclear reactors	○●○○
111. Solar thermal systems use sunlight to produce:	heat	electricity directly	radio waves	sound	●○○○
112. Photovoltaic cells convert sunlight into:	heat only	electricity	sound	chemical energy only	○●○○
113. In a PV cell, photons excite the outer electrons of the semi-metallic element:	silicon	sodium	iron	carbon	●○○○
114. A solar panel produces:	radio waves	alternating current (AC) only	direct current (DC)	no current	○○●○
115. The device that converts DC from solar panels into AC is the:	motor	transformer	generator	inverter	○○○●
116. The basic principle of a PV cell is that when sunlight strikes the material, it:	generates an electric current	breaks the material	emits sound	produces only heat	●○○○

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Questions	A	B	C	D	A B C D
117. The typical lifespan of photovoltaic cells is often:	2 years	25 years or more	1 year	5 years	○●○○
118. An advantage of solar energy is that it is:	expensive to operate	dependent on fuel	renewable and inexhaustible	a source of greenhouse gases	○○●○
119. A disadvantage of solar cells is that:	they need fuel	they produce greenhouse gases	they produce noise	they are not always sunny and output varies	○○○●
120. Solar power does not emit greenhouse gases during:	transport	disposal	electricity generation	manufacturing	○○●○
121. A disadvantage of wind turbines is that they are:	dependent on fuel	polluting	dirty	expensive to set up and wind is not always reliable	○○○●
122. An advantage of wind turbines is that they are:	always reliable	cheap to install	clean and cheap to run	a source of CO_2	○○●○
123. A disadvantage of hydroelectric power stations is that they:	produce radioactive waste	produce greenhouse gases	depend on fuel	are expensive to set up and output may be affected by drought	○○○●
124. An advantage of fossil fuels is that they are:	cheap to build	renewable	clean	reliable	○○○●
125. A disadvantage of fossil fuels is that they:	create pollution and will run out	are inexhaustible	need no refining	are clean	●○○○
126. A disadvantage of nuclear power is that it produces:	acid rain	dangerous radioactive waste	CO_2	photochemical smog	○●○○
127. The best way to meet future energy demands according to the book is to:	stop using electricity	rely only on one source	rely only on fossil fuels	combine energy sources for a balanced and sustainable energy	○○○●
128. To minimise carbon footprint, we should:	burn more fossil fuels	optimise energy usage and adopt renewable energy	drive more	waste electricity	○●○○
129. Which of the following is NOT a strategy to reduce carbon footprint?	Using renewable energy	Efficient transportation	Optimising energy use	Increasing use of fossil fuels	○○○●
130. Alcohol obtained from fermentation of sugarcane and used in gasoline mixtures is:	ethanol	propanol	glycerol	methanol	●○○○
131. Biodiesel is obtained from:	coal	plant oils and animal fats	crude oil only	natural gas	○●○○
132. Ethanol and biodiesel are examples of:	solid fuels	nuclear fuels	fossil fuels	biofuels	○○○●
133. Compressed natural gas (CNG) consists mostly of:	butane	propane	methane	ethane	○○●○

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Questions	A	B	C	D	A B C D
134. LPG consists mainly of propane and:	ethane	butane	methane	pentane	○●○○
135. Diesel is used in:	cooking stoves only	gasoline engines	batteries	diesel engines	○○○●
136. Gasoline is used in:	gasoline (petrol) engines	furnaces only	batteries	diesel engines	●○○○
137. The octane number is a measure of the quality of:	kerosene	bitumen	diesel	petrol	○○○●
138. Catalytic cracking is perfect for making:	bitumen	fuel oil	lubricating oil	petrol (gasoline)	○○○●
139. A student mixes ethanol with petrol to reduce use of fossil fuel. This is an example of using:	a hydrogen fuel	a biofuel blend	solar fuel	nuclear fuel	○●○○
140. A country plans to generate electricity from wind and solar plants to reduce emissions. This is a move toward:	coal power	fossil fuel dependence	nuclear weapons	renewable energy	○○○●
141. A city has many hot sunny days and wants electricity without emissions. The best option is:	diesel generators	coal plants	natural gas plants	solar PV panels	○○○●
142. A coastal region has strong steady winds. The best renewable option is:	wind turbines	nuclear plant	coal plant	oil refinery	●○○○
143. A region has rivers with high water flow throughout the year. A suitable energy source is:	solar thermal	coal	hydropower	oil	○○●○
144. A mountainous country with many rivers builds dams for electricity. A major disadvantage is that:	dams need coal	dams need fuel	dams are expensive to set up and affected by drought	dams produce CO_2	○○●○
145. A refinery wants more petrol from heavy oil fractions. It should use:	steam reforming only	catalytic cracking	bitumen production	fractional distillation only	○●○○
146. A refinery wants to make methanol from natural gas. It should first use:	steam reforming	distillation	thermal cracking	polymerisation	●○○○
147. Fossil fuels are called non-renewable because they:	can be made quickly	are inexhaustible	take millions of years to form and will run out	come from the sun	○○●○
148. The product of complete combustion of a hydrocarbon fuel is carbon dioxide and:	methane	hydrogen	water	oxygen	○○●○
149. The process of splitting a heavy fuel fraction into lighter, more useful fractions is called:	neutralisation	distillation	polymerisation	cracking	○○○●
150. The name given to the gasoline-range products obtained from thermal and catalytic cracking is:	light oil (petrol)	coke	bitumen	lubricating oil	●○○○

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